



CODE OF PRACTICE
DSS/004/040

Specification for Yorkshire Electricity Overhead Lines to CE/C/37(M) and CE/OM/DC1(M)

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Specification for Yorkshire Electricity Overhead Lines to CE/C/37(M) and CE/OM/DC1(M)

DSS/004/040

Written : Ian Scrowston Asset Management	Date : 13/03/00	Signature : I Scrowston
Assured : Tracey Sillito Quality Assurance	Date : 30/11/99	Signature : T H Sillito
Authorised : Reg Dixon System Manager	Date : 16/03/00	Signature : R Dixon

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Revision Record

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1. Purpose

All overhead lines shall meet the requirements of the Electricity Supply Regulations 1988 and the Electricity Supply (Amendment) Regulations 1990. Unlike overhead line regulations previously in force these regulations do not set specific mechanical design requirements, instead they require that

“All suppliers works shall be sufficient for the purpose for, and the circumstances in, which they are used and so constructed, installed, protected, (both electrically and mechanically), used and maintained as to prevent danger or interruption of supply so far as is reasonably practicable”.

Overhead lines, which meet the requirements of this document, comply with the 1988 Supply Regulations, the Electricity Supply (Amendment) Regulations 1990 and any relaxation granted by the Department of Energy.

2. Scope

This specification shall no longer be used for the construction of new lines. It has been updated to reflect changes in the material lists and as such shall only be used for the renovation or replacement of existing structures built to this specification.

Double circuit 33kV and 66kV overhead lines shall be constructed to the specification CE/OM/DC1(M) as detailed in this document.

NOTE: This document is still under amendment by Technical standards with a view to providing a new double circuit line option. Additionally a number extra tables will be inserted to allow its use with Poplar conductor.

3. Lines to CE/C/37(M) and CE/OM/DC1(M)

3.1 Weather Locations

Lines constructed to these specifications are suitable for use in all locations within Yorkshire Electricity's area.

3.2 Conductor

Lines shall be constructed using Aluminium Conductor Steel Reinforced, 175sq.mm.(37/2.79mm.) code name LYNX.

3.3 Span Lengths

Circuits shall be planned such that the average span length of each self supporting section does not exceed the Recommended Span.

For CE/C/37(M) lines longer spans may be obtained by taking advantage of the increased span lengths detailed in the Wind Loading Span table.

3.4 Conductor Erection

Final conductor sags and tensions shall be determined from the Design Sag and Tension Table for each specification.

The erection tension for new conductors shall be determined from the Erection Tension Table for each specification.

Where existing conductors are to be re-used the erection tension shall be determined from the Design Sag and Tension Table for each specification.

3.5 Supports

Supports shall be assembled to the following drawings for CE/C/37(M):-

DRAWING NUMBER	SUPPORT TYPE
Y104 L 0101	Intermediate
Y104 L 0102	Section
Y104 L 0103	Angle < 100
Y104 L 0104	Angle 100 – 600
Y104 L 0105	Slack Span Terminal
Y104 L 0106	33kV Cable Terminal with S.D.
Y104 L 0107	66kV Cable Terminal with S.D.
Y104 L 0108	66kV Cable Terminal with S.D.(tall poles)

and to the following drawings for CE/OM/DC 1(M):-

DRAWING NUMBER	SUPPORT TYPE
Y105 L 0101	Intermediate
Y105 L 0102	Section
Y105 L 0103	Angle

Termination of CE/OM/DC 1(M) shall be on CE/C/37(M) structures.

3.6 Stays

All stay-wires shall be set at 300 or 450 to the pole as determined by the specifications.

Stay insulators shall be installed in accordance with the following table :-

VOLTAGE	EARTHED SUPPORT	UNEARTHED SUPPORT
33kV	1 Type 1 P	2 Type 2 P
66kV	1 Type 1 P	1 Beam Type

NOTE! P = Porcelain

3.7 CE/C/37(M)

175sq.mm ACSR. ALL LEVELS.

Recommended Span. 150m.

Wind Loading Span (m)									
Support Length (m)									

IN LINE STRUCTURES.

DRAWING NUMBER	SUPPORT TYPE	SUPPORT CLASS
Y104 L 0101	H Inter	Stout
Y104 L 0102	H Section	Stout

ANGLE STRUCTURES.

DRAWING NUMBER	SUPPORT TYPE	SUPPORT CLASS	LINE DEVIATION	STAYS No
Y104 L 0103	H Sect Angle	Stout	10 degrees	1 at 30 degrees
Y104 L 0104	H Sect Angle	Stout	18 degrees	1 at 45 degrees
Y104 L 0104	H Sect Angle	Stout	39 degrees	2 at 45 degrees
Y104 L 0104	H Sect Angle	Stout	60 degrees	3 at 45 degrees

Stay Positions.

On 1 or 2 stay installations the stays are fixed to the outer limb of the structure.

On 3 stay installations 1 stay is fixed to the outer limb and 2 stays to the inner limb

TERMINAL STRUCTURES

DRAWING NUMBER	SUPPORT TYPE	SUPPORT CLASS	STAYS No
Y104 L 0105	Slack Span Term (33/66kV)	Stout	2/leg
Y104 L 0106	Cable Term (33/66kV)	Stout	2/leg
Y104 L 0107	Cable Term ≤14m (66kV)	Stout	2/leg
Y104 L 0108	Cable Term >14m (66kV)	Stout	2/leg
Y104 L 0109	Polymeric Cable Termination (33kV)	Stout	2/leg

SAG TABLES

Conductor design sags and tensions are given in table 3.7A

Conductor erection sags and tensions are given in table 3.7B

3.8 CE/OM/DC1(M)

175sq.mm ACSR. ALL LEVELS.

Recommended Span. 120m.

IN LINE STRUCTURES

DRAWING NUMBER	SUPPORT TYPE	SUPPORT CLASS
Y105 L 0101	H Inter	Medium
Y105 L 0102	H Section	Stout

ANGLE STRUCTURES

DRAWING NUMBER	SUPPORT TYPE	SUPPORT CLASS	LINE DEVIATION		STAYS No
			LEVEL GROUND	1 in 10 DOWNPULL	45 degree SLOPE
Y105 L 0103	H Sect Angle	Stout	6	6	1
Y105 L 0103	H Sect Angle	Stout	17	17	2
Y105 L 0103	H Sect Angle	Stout	17	14	2
Y105 L 0103	H Sect Angle	Stout	27	23	3
Y105 L 0103	H Sect Angle	Stout	18	14	3
Y105 L 0103	H Sect Angle	Stout	38	38	4
Y105 L 0103	H Sect Angle	Stout	38	32	4
Y105 L 0103	H Sect Angle	Stout	52	51	5
Y105 L 0103	H Sect Angle	Stout	41	32	5
Y105 L 0103	H Sect Angle	Stout	60	51	6
Y105 L 0103	H Sect Angle	Stout	41	32	6

* Line deviations limited by crippling load of support.

Stay Positions. 1 or 2 stays All outer limb, top crossarm.

3 stays 2 outer limb, top crossarm,
1 outer limb, bottom crossarm.4 stays 1 outer limb, top crossarm,
1 outer limb, bottom crossarm,
2 inner limb, top crossarm.5 stays 2 outer limb, top crossarm,
1 outer limb, bottom crossarm,
2 inner limb, top crossarm.

TERMINAL STRUCTURES

DRAWING NUMBER	SUPPORT TYPE	SUPPORT CLASS	STAYS No
Y104 L 0105	Slack Span Term (33/66kV)	Stout	2/leg
Y104 L 0106	Cable Term (33kV)	Stout	2/leg
Y104 L 0107	Cable Term $\leq 14\text{m}$ (66kV)	Stout	2/leg
Y104 L 0108	Cable Term $> 14\text{m}$ (66kV)	Stout	2/leg
Y104 L 0109	Polymeric Cable Termination (33kV)	Stout	2/leg

SAG TABLES

Conductor design sags and tensions are given in table 3.8A

Conductor erection sags and tensions are given in table 3.8B

4. References

None

5. Definitions

5.1 Auxiliary Equipment

Equipment other than that forming part of the line design which may be placed on supports, such as transformers, fuse-gear etc.

5.2 Average Span

The arithmetic average of a number of spans in a line or self supporting section of line.

5.3 Basic Span

The span length adopted for sag/tension calculations.

5.4 Clashing

This term is used to describe the electrical contact between two line conductors of an overhead line displaced from their normal operating position by environmental forces.

5.5 Conductor Down-pull

The vertical loading imposed by the conductors corresponding to a gradient measured between adjacent points of support.

5.6 Factor of Safety

The ratio of the failing stress of an overhead line component to the designed maximum working stress of that component

5.7 Freezing Point Tension

The design tension of a conductor unloaded at 00C.

5.8 Intermediate Support

A support in a straight run of a line on which the conductors are supported on pin, line post or suspension insulators.

5.9 Maximum Conductor Tension

The likely maximum conductor tension assessed at 00C which is limited by either the application of the Maximum Conductor Resultant or of a vibration limit.

5.10 Maximum Clashing Span

The maximum span length approved for each site, conductor and crossarm combination.

5.11 Over-tensioning

Excess tension applied above normal tension at time of erection.

5.12 Pin Angle Support:-

A support at which a line deviates and the conductors are supported on pin or line post insulators.

5.13 Recommended Span

The average span length in any section to which the line should be planned.

5.14 Sag

The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured mid-span.

5.15 Section Angle Support

A support at which a line deviates and the conductors are made off on either side of the crossarm on tension insulator sets.

5.16 Section Support

A support in the straight run of a line on which the conductors are made off on either side of the crossarm on tension insulator sets.

5.17 Self Supporting Section

Any section of overhead line between supports stayed for terminal loading.

5.18 Span

The horizontal distance between adjacent supports.

5.19 Slack Span

The span of conductors between a terminal support and a supporting structure in a substation or an adjacent overhead line.

5.20 Wind Loading Span

The wind loading span associated with any support is half the sum of the two adjacent spans.

5.21 Weight Span

That span of conductor, the weight of which corresponds to the vertical load imposed by a conductor on a given support.

Appendix 1**DESIGN SAGS AND TENSIONS. 175mm². ACSR. ALL LEVELS.
150m basic span.**

Conductor Breaking Load. 79800 Newtons.

Conductor Diameter. 19.53mm.

Conductor Weight. 0.8628 kg/m.

Conductor c.s.a. 226.2 sq.mm.

Maximum Working Tension. 25740 Newtons.

Span (m)	Temp. degrees C															
	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75
60	0.23	0.25	0.27	0.28	0.30	0.32	0.34	0.35	0.37	0.39	0.41	0.42	0.44	0.46	0.48	
65	0.28	0.29	0.31	0.33	0.35	0.37	0.39	0.41	0.44	0.46	0.48	0.50	0.52	0.54	0.56	0.58
70	0.32	0.34	0.36	0.38	0.41	0.43	0.46	0.48	0.51	0.53	0.55	0.58	0.60	0.62	0.65	0.67
75	0.37	0.39	0.42	0.44	0.47	0.50	0.52	0.55	0.58	0.61	0.64	0.66	0.69	0.72	0.74	0.77
80	0.42	0.44	0.47	0.50	0.53	0.56	0.60	0.63	0.66	0.69	0.72	0.75	0.79	0.82	0.85	0.88
85	0.47	0.50	0.53	0.57	0.60	0.64	0.67	0.71	0.75	0.78	0.82	0.85	0.89	0.92	0.95	0.99
90	0.53	0.56	0.60	0.64	0.67	0.71	0.75	0.79	0.84	0.88	0.92	0.96	0.99	1.03	1.07	1.11
95	0.59	0.63	0.67	0.71	0.75	0.80	0.84	0.89	0.93	0.98	1.02	1.06	1.11	1.15	1.19	1.23
100	0.65	0.69	0.74	0.79	0.83	0.88	0.93	0.98	1.03	1.08	1.13	1.18	1.23	1.27	1.32	1.37
105	0.72	0.77	0.81	0.87	0.92	0.97	1.03	1.08	1.14	1.19	1.25	1.30	1.35	1.41	1.46	1.51
110	0.79	0.84	0.89	0.95	1.01	1.07	1.13	1.19	1.25	1.31	1.37	1.43	1.49	1.54	1.60	1.65
115	0.86	0.92	0.98	1.04	1.10	1.17	1.23	1.30	1.36	1.43	1.49	1.56	1.62	1.69	1.75	1.81
120	0.94	1.00	1.06	1.13	1.20	1.27	1.34	1.41	1.49	1.56	1.63	1.70	1.77	1.84	1.90	1.97
125	1.02	1.09	1.15	1.23	1.30	1.38	1.46	1.53	1.61	1.69	1.77	1.84	1.92	1.99	2.07	2.14
130	1.10	1.17	1.25	1.33	1.41	1.49	1.57	1.66	1.74	1.83	1.91	1.99	2.07	2.15	2.23	2.31
135	1.19	1.27	1.35	1.43	1.52	1.61	1.70	1.79	1.88	1.97	2.06	2.15	2.24	2.32	2.41	2.49
140	1.28	1.36	1.45	1.54	1.63	1.73	1.93	1.92	2.02	2.12	2.22	2.31	2.41	2.50	2.59	2.68
145	1.37	1.46	1.55	1.65	1.75	1.85	1.96	2.06	2.17	2.27	2.38	2.48	2.58	2.68	2.78	2.88
150	1.47	1.56	1.66	1.77	1.87	1.98	2.10	2.21	2.32	2.43	2.54	2.65	2.76	2.87	2.97	3.08
155	1.57	1.67	1.78	1.89	2.00	2.12	2.24	2.36	2.48	2.60	2.72	2.83	2.95	3.06	3.18	
160	1.67	1.78	1.89	2.01	2.13	2.26	2.38	2.51	2.64	2.77	2.89	3.02	3.14	3.26	3.38	3.50
165	1.78	1.89	2.01	2.14	2.27	2.40	2.54	2.67	2.81	2.94	3.08	3.21	3.34	3.47	3.60	3.72
170	1.88	2.01	2.14	2.27	2.41	2.55	2.69	2.84	2.98	3.12	3.27	3.41	3.55	3.68	3.82	3.95
175	2.00	2.13	2.26	2.41	2.55	2.70	2.85	3.01	3.16	3.31	3.46	3.61	3.76	3.90	4.05	4.19
180	2.11	2.25	2.39	2.54	2.70	2.86	3.02	3.18	3.34	3.50	3.66	3.82	3.98	4.13	4.28	4.43
Tension at basic span																
(N)	16264	15269	14348	13501	12726	12022	11383	10804	10282	9809	9381	8992	8639	8317	8023	7754
(kgf)	1658	1557	1463	1377	1298	1226	1161	1102	1048	1000	957	917	881	848	818	791
lbf	3653	3430	3223	3033	2859	2700	2557	2427	2310	2203	2107	2020	1941	1868	1802	1742

Appendix 2**Erection sag and tension table. 175mm². ACSR. All levels.
150m basic span.**

Conductor Breaking Load. 79800 Newtons.

Conductor Diameter. 19.53mm.

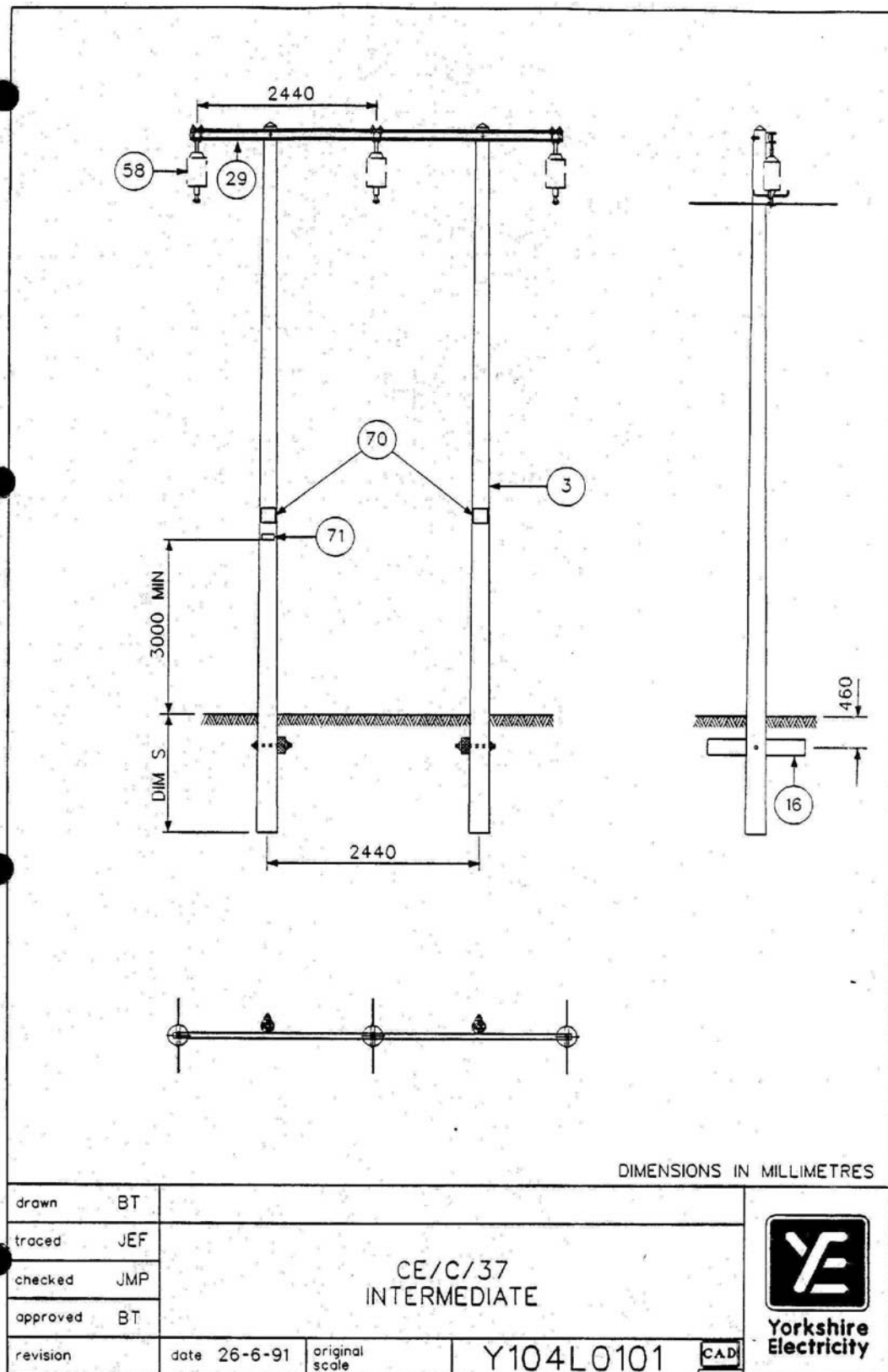
Conductor Weight. 0.8628 kg/m.

Conductor c.s.a. 226.2 mm².

Maximum Working Tension. 25740 Newtons.

Span (m)	Temp. degrees C															
	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75
60	0.22	0.23	0.25	0.26	0.28	0.30	0.31	0.33	0.35	0.37	0.39	0.40	0.42	0.44	0.46	0.47
65	0.25	0.27	0.29	0.31	0.33	0.35	0.37	0.39	0.41	0.43	0.45	0.47	0.49	0.51	0.53	0.55
70	0.30	0.32	0.34	0.36	0.38	0.40	0.43	0.45	0.48	0.50	0.52	0.55	0.57	0.60	0.62	0.64
75	0.34	0.36	0.39	0.41	0.44	0.46	0.49	0.52	0.55	0.57	0.60	0.63	0.66	0.68	0.71	0.74
80	0.39	0.41	0.44	0.47	0.50	0.53	0.56	0.5	0.62	0.65	0.69	0.72	0.75	0.78	0.81	0.84
85	0.44	0.46	0.49	0.53	0.56	0.59	0.63	0.67	0.70	0.74	0.77	0.81	0.84	0.88	0.91	0.95
90	0.49	0.52	0.55	0.59	0.63	0.67	0.71	0.75	0.79	0.83	0.87	0.91	0.95	0.99	1.02	1.06
95	0.54	0.58	0.62	0.66	0.70	0.74	0.79	0.83	0.88	0.92	0.97	1.01	1.05	1.10	1.14	1.18
100	0.60	0.64	0.69	0.73	0.78	0.82	0.87	0.92	0.97	1.02	1.07	1.12	1.17	1.22	1.26	1.31
105	0.67	0.71	0.76	0.80	0.85	0.91	0.96	1.02	1.07	1.13	1.18	1.23	1.29	1.34	1.39	1.45
110	0.73	0.78	0.83	0.88	0.94	1.00	1.05	1.11	1.17	1.23	1.30	1.35	1.41	1.47	1.53	1.59
115	0.80	0.85	0.91	0.96	1.03	1.09	1.15	1.22	1.28	1.35	1.42	1.48	1.55	1.61	1.67	1.73
120	0.87	0.93	0.99	1.03	1.12	1.18	1.25	1.33	1.40	1.47	1.54	1.61	1.68	1.75	1.82	1.89
125	0.94	1.00	1.07	1.14	1.21	1.29	1.36	1.44	1.52	1.59	1.67	1.75	1.83	1.90	1.98	2.05
130	1.02	1.09	1.16	1.23	1.31	1.39	1.47	1.56	1.64	1.72	1.81	1.89	1.98	2.06	2.14	2.22
135	1.10	1.17	1.25	1.33	1.41	1.50	1.59	1.68	1.77	1.86	1.95	2.04	2.13	2.22	2.30	2.39
140	1.18	1.26	1.34	1.43	1.52	1.61	1.71	1.80	1.90	2.00	2.10	2.19	2.29	2.39	2.48	2.57
145	1.27	1.35	1.44	1.53	1.63	1.73	1.83	1.94	2.04	2.15	2.25	2.35	2.46	2.56	2.66	2.76
150	1.36	1.45	1.54	1.64	1.74	1.85	1.96	2.07	2.18	2.30	2.41	2.52	2.63	2.74	2.85	2.95
155	1.45	1.54	1.65	1.75	1.86	1.98	2.09	2.21	2.33	2.45	2.57	2.69	2.81	2.92	3.04	3.15
160	1.54	1.65	1.75	1.87	1.98	2.11	2.23	2.36	2.48	2.61	2.74	2.87	2.99	3.12	3.24	3.36
165	1.64	1.75	1.87	1.99	2.11	2.24	2.37	2.51	2.64	2.78	2.91	3.05	3.18	3.31	3.44	3.57
170	1.74	1.86	1.98	2.11	2.24	2.38	2.52	2.66	2.80	2.95	3.09	3.24	3.38	3.52	3.65	3.73
175	1.85	1.97	2.10	2.23	2.37	2.52	2.67	2.82	2.97	3.13	3.28	3.43	3.58	3.73	3.87	4.02
180	1.96	2.08	2.22	2.36	2.51	2.67	2.82	2.98	3.14	3.31	3.47	3.63	3.79	3.94	4.10	4.32
Tension at basic span																
(N)	17570	16490	15479	14542	13678	12888	12169	11516	10925	10391	99.8	9470	9074	8713	8385	8085
(kgf)	1792	16/81	1578	1483	1395	1314	1241	1174	1114	1060	1010	966	925	888	855	824
lbf	3947	3704	3477	3267	3073	2895	2734	2587	2454	2334	2226	2127	2038	1957	1884	1816

Appendix 3
Drawing Y104L0101



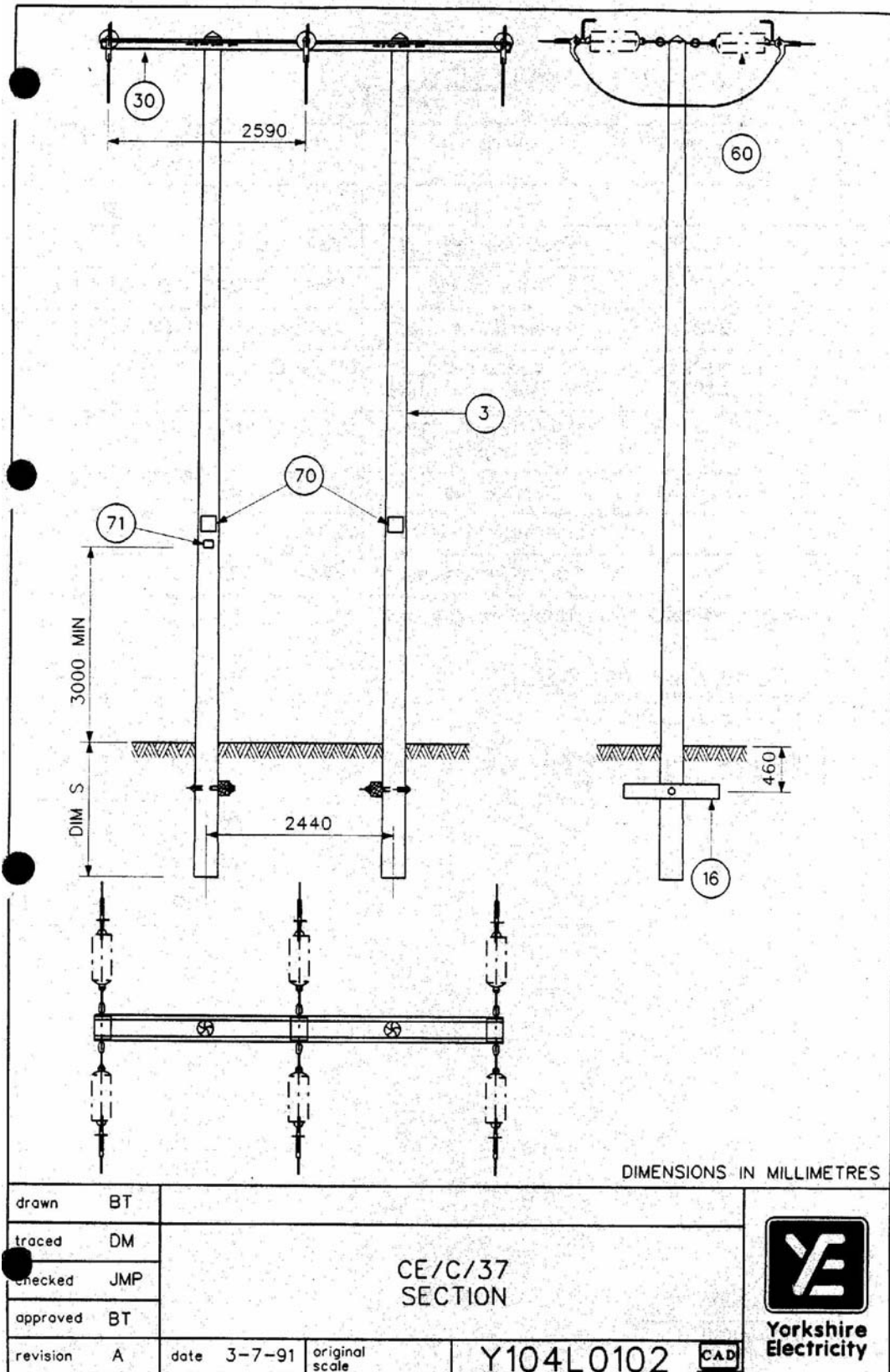
Appendix 4**Materials List for Y104L0101**

Item	Component	Drawing No	Quantity	Cat No.
Pole & Foundations				
3	Wood Pole (Type 1) (Med or Stout to suit spanning)	Y204 L0601	2	To Suit Pole length & span
16	Kicking / Stay Block (Type 2)	439103	2	234476
16	Bolts, M20 x 530 Galvd		2	372815
16	Washers, Square Curved	439605	2	368078
16	Washer, Square Flat	0183/3	2	368063
Crossarms				
29	Crossarm Member	Y204 L0901	1	222844
29	Washers, Square Curved	439605	2	368078
29	Washer, Square Flat	0183/3	2	368063
29	Bolts, M20 x 300 (Med or Stout)	-	2	378794
Insulator Assemblies				
58	* 33 or 66kV Suspension Set	1091010488 sht 6	3	See NSP/001/027
Sign & Notices				
70	Safety Sign	1091010229 sht1	2	363318
70 / 71	Fibre Washers, 14mm x 6mm thick	Spacer washers placed under safety sign	10	374581
70 / 71	Fibre Washers, 14mm x 2mm thin	Washers placed on top of sign	10	374555
70 / 71	Brass Slotted screws No1		10	375777
71	Number base plate	1091010028 sht3	1	243258
71	Pole Numbers	1091010028 sht3	To Suit	To Suit

* Detailed Insulator information now available within insulator guidance document

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0-13.0	2.00
14.0-15.0	2.10
16.0-17.0	2.25
18.0-20.0	2.40

Appendix 5
Drawing Y104L0102



Appendix 6

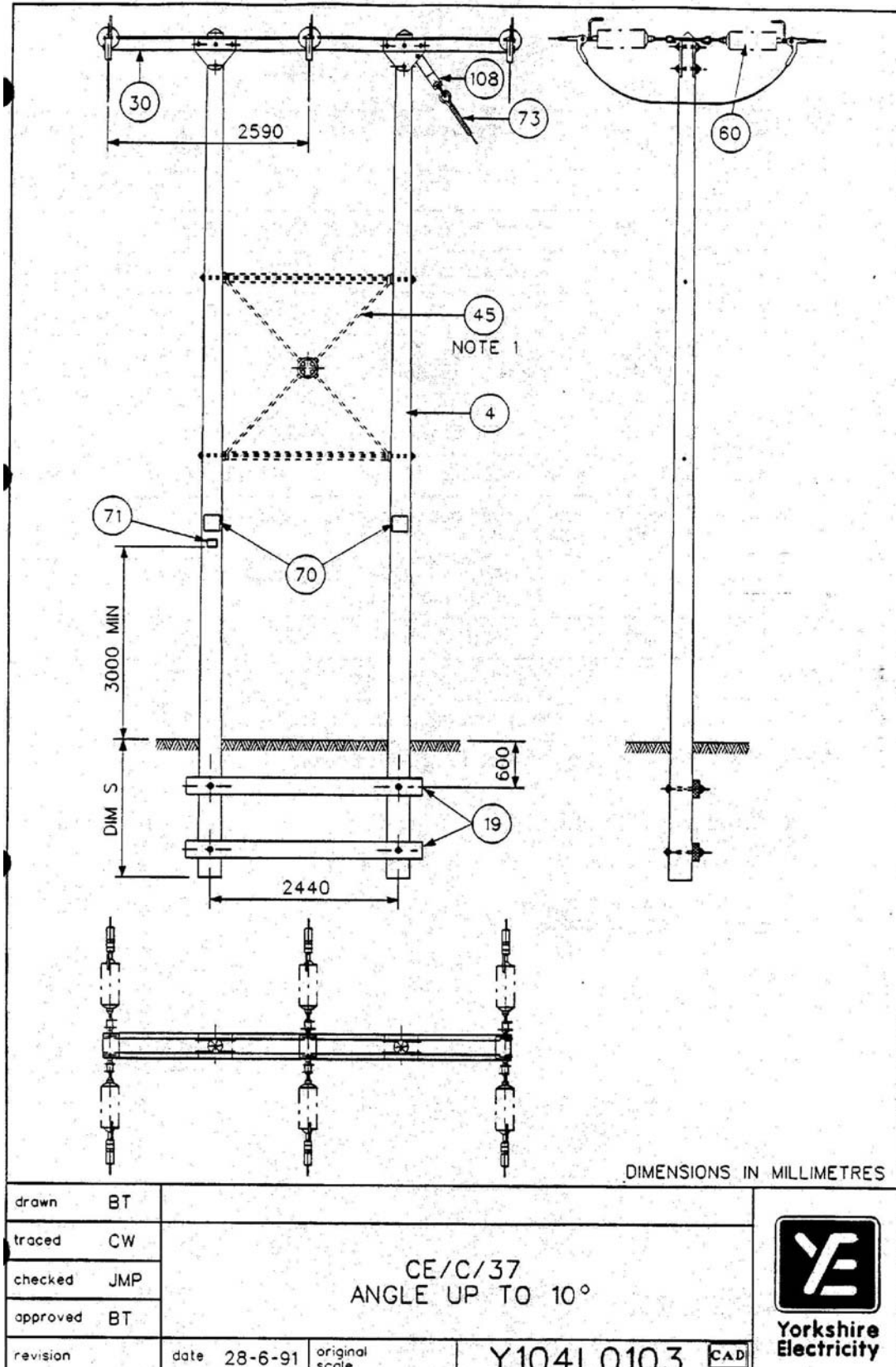
Materials List for Y104L0102

Item	Component	Drawing No	Quantity	Cat No.
Pole & Foundations				
3	Wood Pole (Type 1) (Stout)	Y204 L0601	2	To Suit Pole
16	Kicking / Stay Block (Type 2)	439103	2	234476
16	Bolts, M20 x 530 Galvd		2	372815
16	Washers, Square Curved	439605	2	368078
16	Washer, Square Flat	0183/3	2	368063
Crossarms				
29	Crossarm Member	Y204 L0902	2	222455
29	Washer, Square Flat	0183/3	2	368063
29	Bolts, M20 x 300 (Med or Stout)	-	2	378794
Insulator Assemblies				
60	* 33 or 66kV Tension Set	1091010488 sht 9	6	See NSP/001/027
Sign & Notices				
70	Safety Sign	1091010229 sht1	2	363318
70 / 71	Fibre Washers, 14mm x 6mm thick	Spacer washers placed under safety sign	10	374581
70 / 71	Fibre Washers, 14mm x 2mm thin	Washers placed on top of sign	10	374555
70 / 71	Brass Slotted screws No1		10	375777
71	Number base plate	1091010028 sht3	1	243258
71	Pole Numbers	1091010028 sht3	To Suit	To Suit

* Detailed Insulator information now available within insulator guidance document

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0-13.0	2.00
14.0-15.0	2.10
16.0-17.0	2.25
18.0-20.0	2.40

Appendix 7
Drawing Y104L0103



Appendix 8

Materials List for Y104L0103

Item	Component	Drawing No	Quantity	Cat No.
Pole & Foundations				
4	Wood Pole (Type 2) (Stout)	Y204 L0601	2	To Suit Pole
19	Brace Block (2.44m Centres)	Y204 L0301	2	217497
19	Bolts, M20 x 530 Galvd		4	372815
19	Washers, Square Curved	439605	4	368078
19	Washer, Square Flat	0183/3	4	368063
45	Pole Bracing Assembly (Single)	Y204 L0905	1	219519
45	Pole Bracing Assembly (Double)	Y204 L0905	1	219501
Crossarms				
30	Crossarm Member, Conductor Plates & Stay Gusset Plates	Y204 L0902	1 Set	222455
30	Bolts, M20 x 60 bolts (Gusset Plates)		8	378703
30	Washer, Square Flat	0183/3	4	368063
30	Bolts, M20 x 300 (Med or Stout)	-	4	378794
Insulator Assemblies				
60	* 33 or 66kV Tension Set	1091010488 sht 9	6	See NSP/001/027
Sign & Notices				
70	Safety Sign	1091010229 sht1	2	363318
70 / 71	Fibre Washers, 14mm x 6mm thick	Spacer washers placed under safety sign	10	374581
70 / 71	Fibre Washers, 14mm x 2mm thin	Washers placed on top of sign	10	374555
70 / 71	Brass Slotted screws No1		10	375777
71	Number base plate	1091010028 sht3	1	243258
71	Pole Numbers	1091010028 sht3	To Suit	To Suit
Stays				
73	See NSP/001/104 Guidance document for stays Assemblies	Y204 L0501 &	To Suit	See document for details
73	** 120kN Insulator Stay Assembly	1 09 123 0372 Sht2	To Suit	251547

* Detailed Insulator information now available within insulator guidance document

** Updated stay assembly

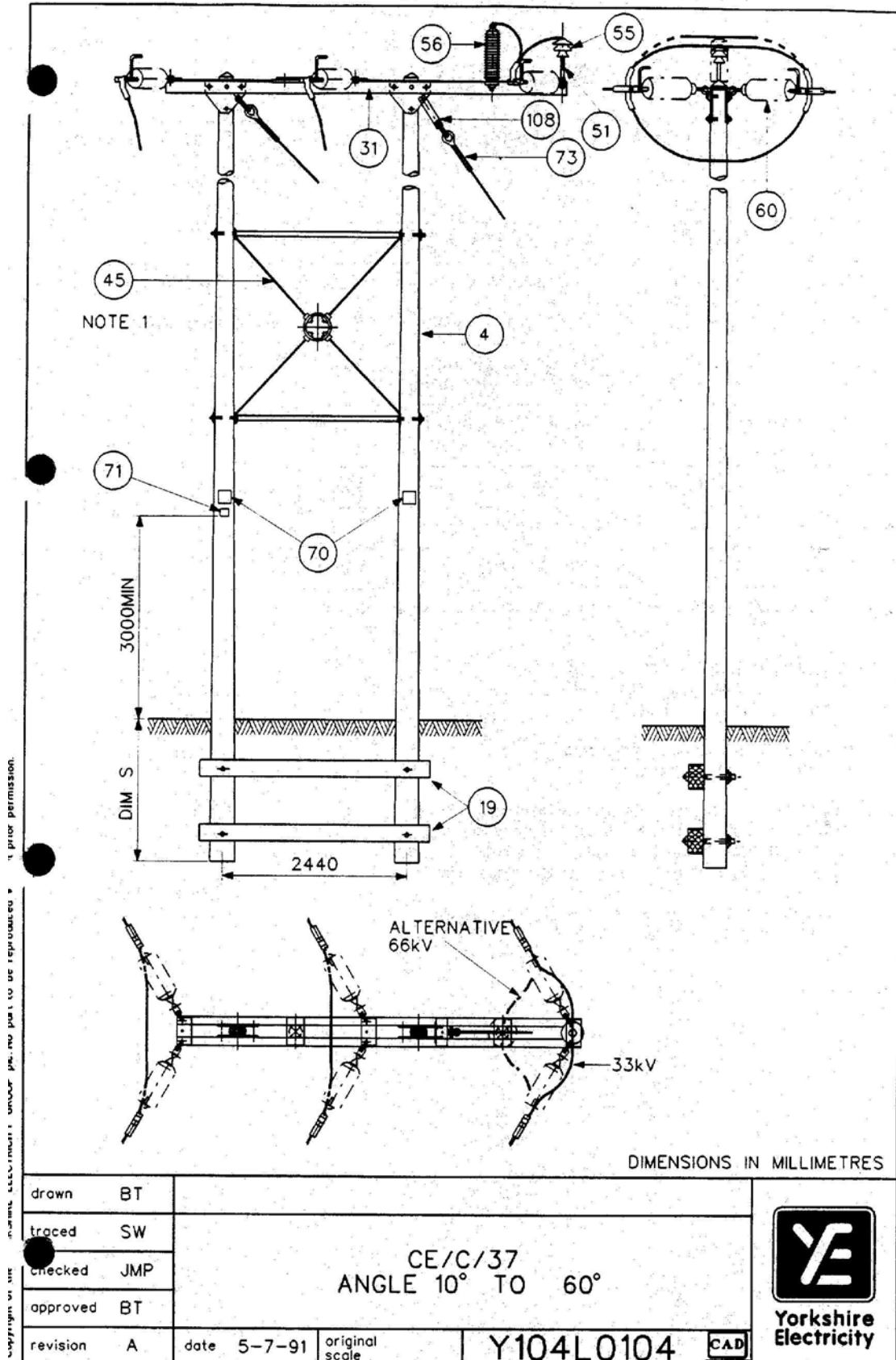
Notes to drawing No Y104 L0103

- Single pole bracing required for poles up to 14.0m.

Double pole bracing required for poles 15.0m and above.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0-13.0	2.00
14.0-15.0	2.10
16.0-17.0	2.25
18.0-20.0	2.40

Appendix 9
Drawing Y104L0104



Appendix 10**Materials List for Y104L0104**

Item	Component	Drawing No	Quantity	Cat No.
Pole & Foundations				
4	Wood Pole (Type 2) (Stout)	Y204 L0601	2	To Suit
19	Brace Block (2.44m Centres)	Y204 L0301	2	217497
19	Bolts, M20 x 530 Galvd		4	372815
19	Washers, Square Curved	439605	4	368078
19	Washer, Square Flat	0183/3	4	368063
45	Pole Bracing Assembly (Single)	Y204 L0905	1	219519
45	Pole Bracing Assembly (Double)	Y204 L0905	1	219501
Crossarms				
31	Crossarm Member, Conductor Plates & Stay Gusset Plates	Y204 L0903	1 Set	222463
31	Bolts, M20 x 60 bolts (Gusset Plates)		8	378703
31	Washer, Square Flat	0183/3	4	368063
31	Bolts, M20 x 300 (Med or Stout)	-	4	378794
Insulator Assemblies				
60	* 33 or 66kV Tension Set	1091010488 sht 9	6	See NSP/001/027
51	Pilot Pin	1091010619/1 or BS3288 fig5 ref 50	1	261909
55	33kv Pin Insulator	1091010486 sht 4 or 9	1	253423
56	66kv Post Insulator	1091010487 sht 12	1	253702
Sign & Notices				
70	Safety Sign	1091010229 sht1	2	363318
70 / 71	Fibre Washers, 14mm x 6mm thick	Spacer washers placed under safety sign	10	374581
70 / 71	Fibre Washers, 14mm x 2mm thin	Washers placed on top of sign	10	374555
70 / 71	Brass Slotted screws No1		10	375777
71	Number base plate	1091010028 sht3	1	243258
71	Pole Numbers	1091010028 sht3	To Suit	To Suit
Stays				
73	See NSP/001/104 Guidance document for stays Assemblies	Y204 L0501 &	To Suit	See document for details
73	** 120kN Insulator Stay Assembly	1 09 123 0372 Sht2	To Suit	251547

* Detailed Insulator information now available within insulator guidance document

** Updated stay assembly

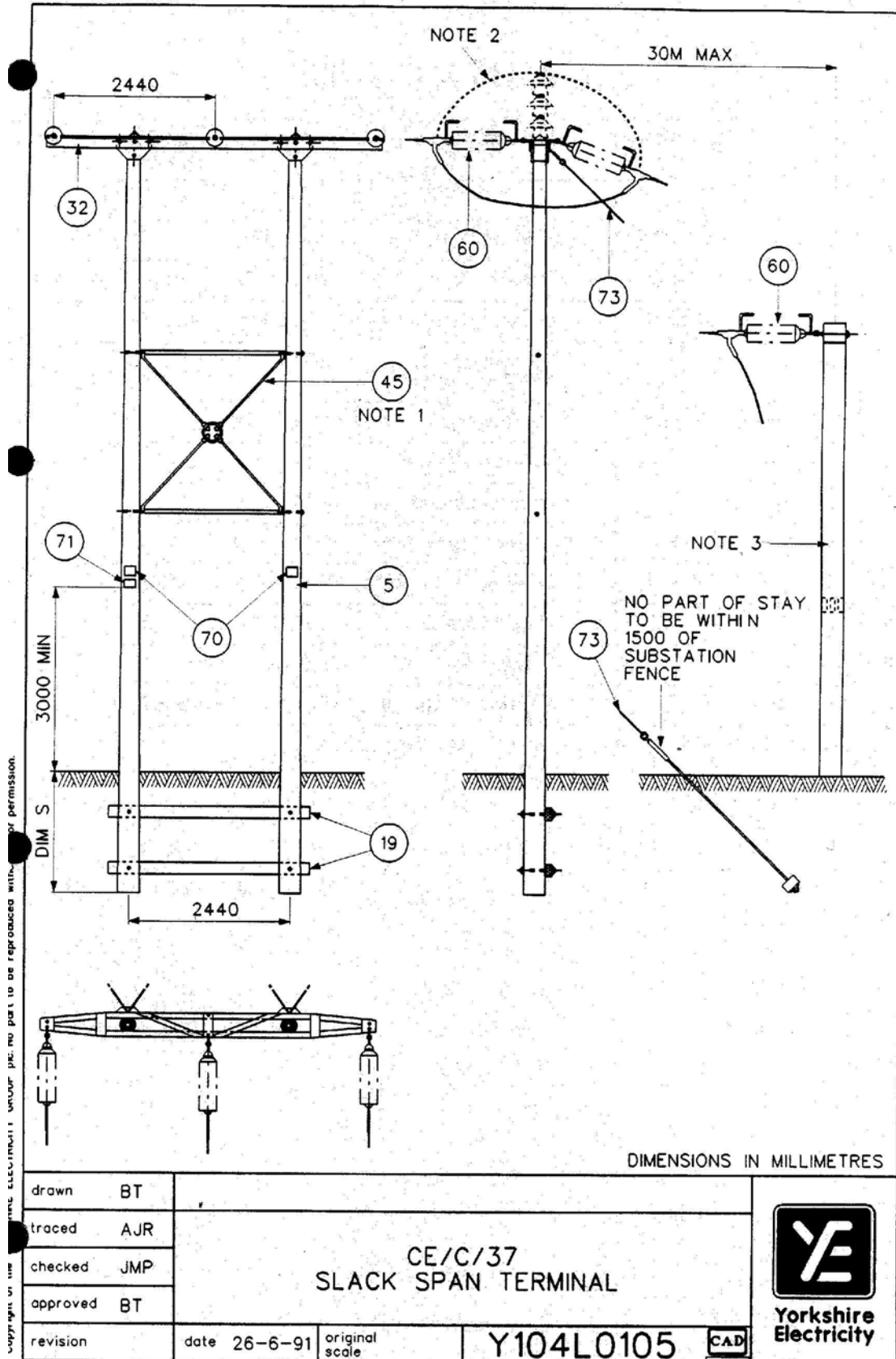
Notes to drawing No Y104 L0104

- Single pole bracing required for poles up to 14.0m.
Double pole bracing required for poles 15.0m and above.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0-13.0	2.00
14.0-15.0	2.10
16.0-17.0	2.25
18.0-20.0	2.40

Appendix 11

Drawing Y104L0105



Appendix 12

Materials List for Y104L0105

Item	Component	Drawing No	Quantity	Cat No.
Pole & Foundations				
5	Wood Pole (Stout)	Y204 L0601	2	To Suit Pole
19	Brace Block (2.44m Centres)	Y204 L0301	2	217497
19	Bolts, M20 x 530 Galvd		4	372815
19	Washers, Square Curved	439605	4	368078
19	Washer, Square Flat	0183/3	4	368063
45	Pole Bracing Assembly (Single)	Y204 L0905	1	219519
45	Pole Bracing Assembly (Double)	Y204 L0905	1	219501
Crossarms				
32	Crossarm Member, Conductor Plates & Stay Gusset Plates	Y204 L0904	1 Set	223669
32	Bolts, M20 x 60 bolts (Gusset Plates)		8	378703
32	Washer, Square Flat	0183/3	4	368063
32	Bolts, M20 x 300 (Med or Stout)	-	4	378794
Insulator Assemblies				
60	* 33 or 66kV Tension Set	1091010488 sht 9	6	See NSP/001/027
51	Pilot Pin	1091010619/1 or BS3288 fig5 ref 50	1	261909
55	33kv Pin Insulator	1091010486 sht 4 or 9	1	253423
56	66kv Post Insulator	1091010487 sht 12	1	253702
Sign & Notices				
70	Safety Sign	1091010229 sht1	2	363318
70 / 71	Fibre Washers, 14mm x 6mm thick	Spacer washers placed under safety sign	10	374581
70 / 71	Fibre Washers, 14mm x 2mm thin	Washers placed on top of sign	10	374555
70 / 71	Brass Slotted screws No1		10	375777
71	Number base plate	1091010028 sht3	1	243258
71	Pole Numbers	1091010028 sht3	To Suit	To Suit
Stays				
73	See NSP/001/104 Guidance document for stays Assemblies	Y204 L0501 &	To Suit	See document for details
73	** 120kN Insulator Stay Assembly	1 09 123 0372 Sht2	To Suit	251547

* Detailed Insulator information now available within insulator guidance document

** Updated stay assembly

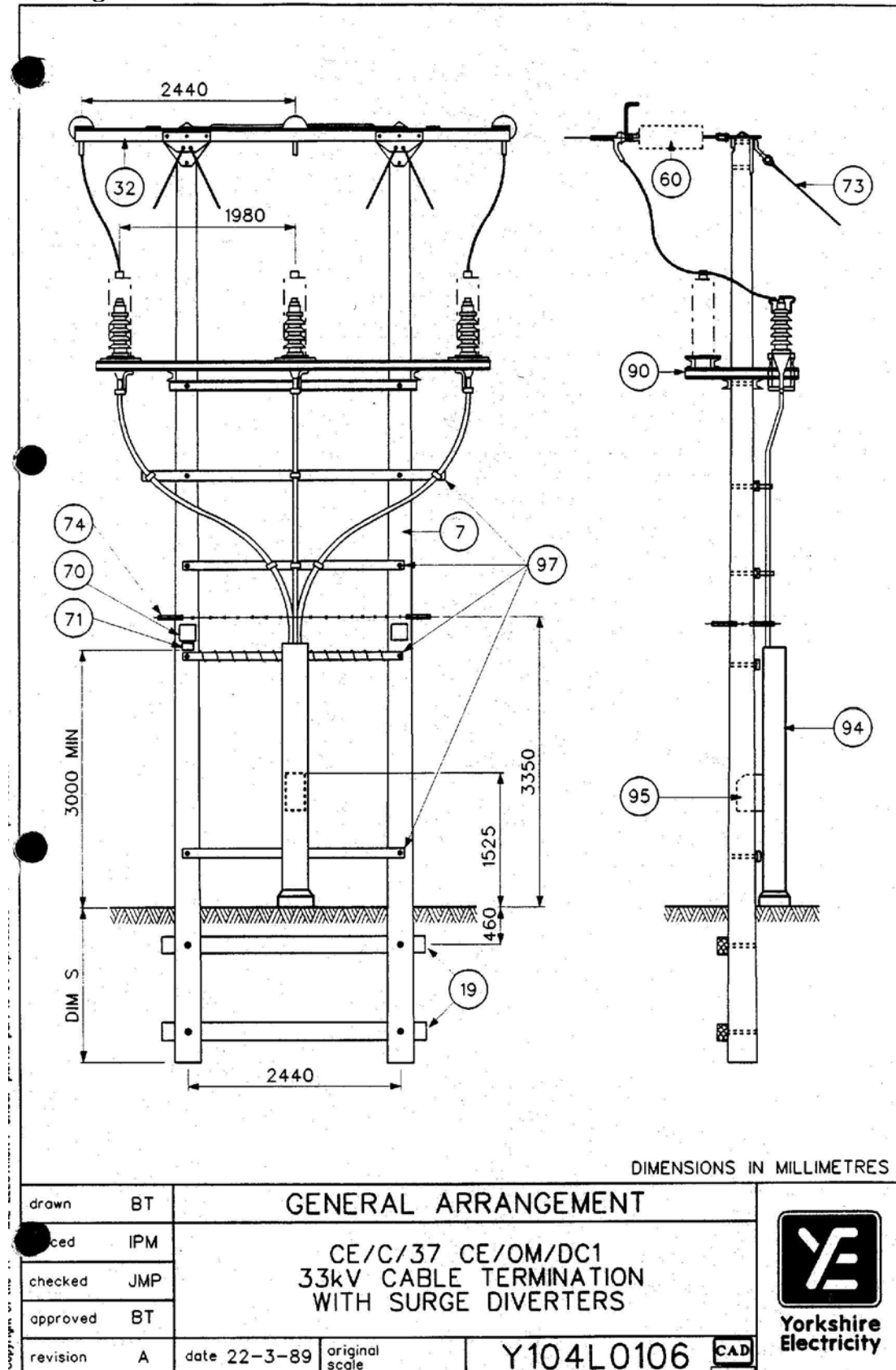
Notes to drawing No Y104 L0105

- Single pole bracing required for poles up to 14.0m.
Double pole bracing required for poles 15.0m and above.
- Alternative jumper position using 66kV post insulator, or 33kV pin or line post insulator, to be used where adequate clearances cannot be obtained with underslung jumper position.

Arrangement of concrete slack span support to drawing number Y003 S3307.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0-13.0	2.00
14.0-15.0	2.10
16.0-17.0	2.25
18.0-20.0	2.40

Appendix 13
Drawing Y104L0106



Appendix 14**Materials List for Y104L0106**

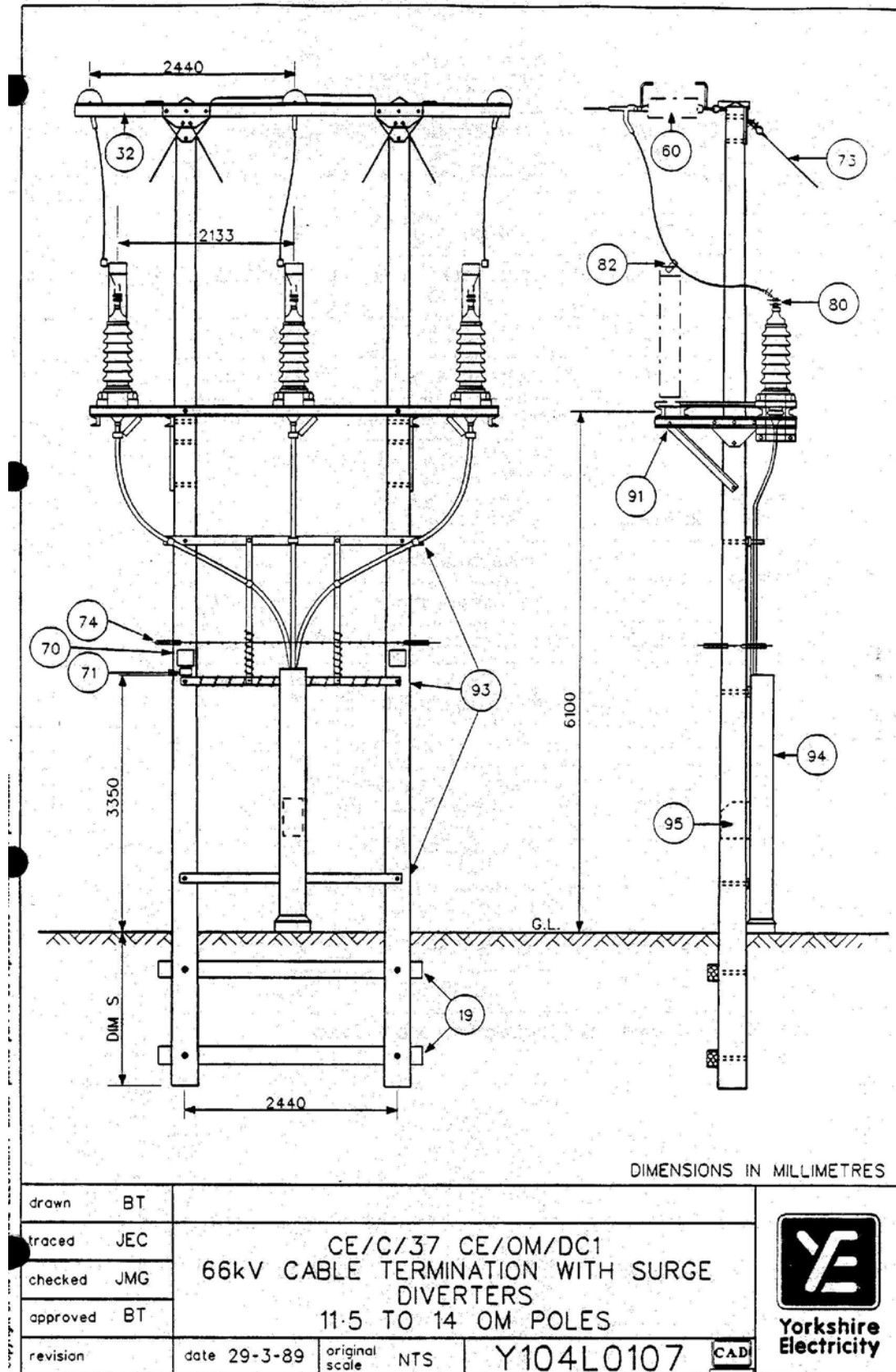
Item	Component	Drawing No	Quantity	Cat No.
Pole & Foundations				
7	Wood Pole (Type 3) (Stout)	Y204 L0602	2	To Suit Pole
19	Brace Block (2.44m Centres)	Y204 L0301	2	217497
19	Bolts, M20 x 530 Galvd		4	372815
19	Washers, Square Curved	439605	4	368078
19	Washer, Square Flat	0183/3	4	368063
Crossarms				
32	Crossarm Member, Conductor Plates & Stay Gusset Plates	Y204 L0904	1 Set	223669
32	Bolts, M20 x 60 bolts (Gusset Plates)		8	378703
32	Washer, Square Flat	0183/3	4	368063
32	Bolts, M20 x 300 (Med or Stout)	-	4	378794
Insulator Assemblies				
60	* 33kV Tension Set	1091010488 sht 9	6	See NSP/001/027
51	Pilot Pin	1091010619/1 or BS3288 fig5 ref 50	1	261909
55	33kv Pin Insulator	1091010486 sht 4 or 9	1	253423
Sign & Notices				
70	Safety Sign	1091010229 sht1	2	363318
70 / 71	Fibre Washers, 14mm x 6mm thick		10	374581
70 / 71	Fibre Washers, 14mm x 2mm thin		10	374555
70 / 71	Brass Slotted screws No1		10	375777
71	Number base plate	1091010028 sht3	1	243258
71	Pole Numbers	1091010028 sht3	To Suit	To Suit
Stays				
73	See NSP/001/104 Guidance document for stays Assemblies	Y204 L0501 &	4	See document for details
73	** 120kN Insulator Stay Assembly	1 09 123 0372 Sht2	4	251547
Cable Support				
74	Anti-Climbing Guard	Y201 L1101/2	1	224519
90	33kV Sealing End P-Form	Y204 L0907	1	227173
	33kV Surge Diverter to be supplied with insulated base		3	246969
94	Cable Guard	Y204 L0911	1	224535
94	Bolts, M20 x 60 bolts		4	378703
95	Link Box	Eng Rec C55		To Suit
97	33kV Cable Supports	Y204 L0915	8	
Earthing				
	70mm (7/3.55)HD CU Cond. PVC Covered type 8 earthing and bonding lead		20m Approx.	220146
	Comp Lug – M22 Hole (70mm PVC Cu)	1091010237	As Req	075234
	Polyprop cleats		As Req	248815
	Earthing Lead casing		1	235548
	Earth Rod	1091013143 sht 1	As Req	354249
	Saddle Clamp	1091013143 sht 2	As Req	354388

* Detailed Insulator information now available within insulator guidance document

** Updated stay assembly

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0-13.0	2.00
14.0-15.0	2.10
16.0-17.0	2.25
18.0-19.0	2.40

Appendix 15
Drawing Y104L0107



Appendix 16**Materials List for Y104L0107**

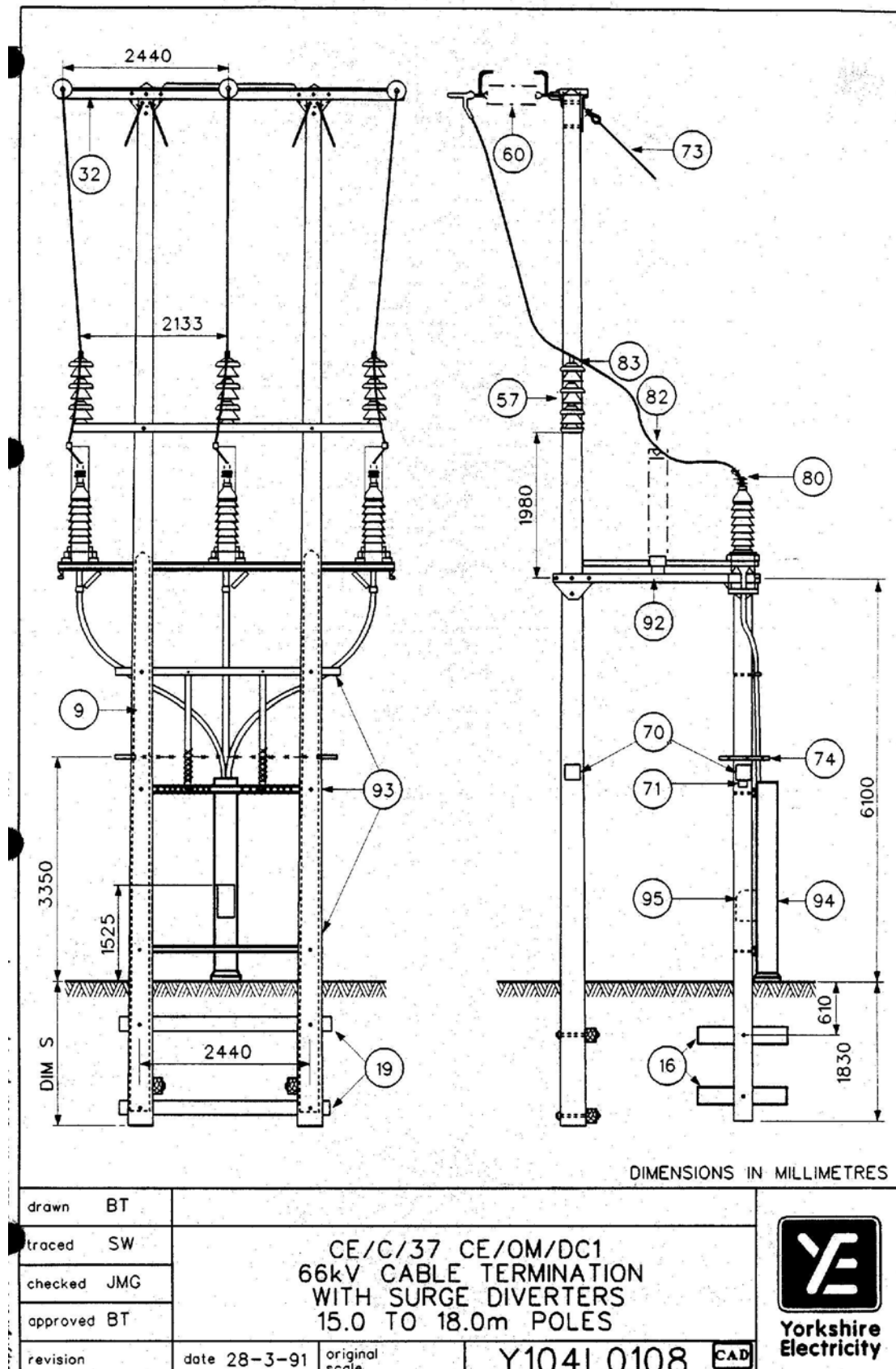
Item	Component	Drawing No	Quantity	Cat No.
	Pole & Foundations			
8	Wood Pole (Stout)	Y204 L0603	2	To Suit Pole
19	Brace Block (2.44m Centres)	Y204 L0301	2	217497
19	Bolts, M20 x 530 Galv		4	372815
19	Washers, Square Curved	439605	4	368078
19	Washer, Square Flat	0183/3	4	368063
	Crossarms			
32	Crossarm Member, Conductor Plates & Stay Gusset Plates	Y204 L0904	1 Set	223669
32	Bolts, M20 x 60 bolts (Gusset Plates)		8	378703
32	Washer, Square Flat	0183/3	4	368063
32	Bolts, M20 x 300 (Med or Stout)	-	4	378794
	Insulator Assemblies			
60	* 66kV Tension Set	1091010488 sht 9	6	See NSP/001/027
	Sign & Notices			
70	Safety Sign	1091010229 sht1	2	363318
70 / 71	Fibre Washers, 14mm x 6mm thick		10	374581
70 / 71	Fibre Washers, 14mm x 2mm thin		10	374555
70 / 71	Brass Slotted screws No1		10	375777
71	Number base plate	1091010028 sht3	1	243258
71	Pole Numbers	1091010028 sht3	To Suit	To Suit
	Stays			
73	See NSP/001/104 Guidance document for stays Assemblies	Y204 L0501 &	4	See document for details
73	** 120kN Insulator Stay Assembly	1 09 123 0372 Sht2	4	251547
	Cable Support			
74	Anti-Climbing Guard	Y201 L1101/2	1	224519
80	66kV Sealing End Connector	Y204 L0801	3	221119
91	66kV Sealing End P-Form	Y204 L0908	1	227116
82	66kV Surge Diverter to be supplied with insulated base		3	224154
82	Surge Diverter Connector	Y204 L0803	3	221119
94	Cable Guard	Y204 L0911	1	224535
94	Bolts, M20 x 60 bolts		14	378703
95	Link Box	Eng Rec C55		To Suit
97	66kV Cable Supports	Y204 L0910	6	
	Earthing			
	70mm (7/3.55)HD CU Cond. PVC Covered type 8 earthing and bonding lead		20m Approx.	220146
	Comp Lug – M22 Hole (70mm PVC Cu)	1091010237	As Req	075234
	Polyprop cleats		As Req	248815
	Earthing Lead casing		1	235548
	Earth Rod	1091013143 sht 1	As Req	354249
	Saddle Clamp	1091013143 sht 2	As Req	354388

* Detailed Insulator information now available within insulator guidance document

** Updated stay assembly

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0-13.0	2.00
14.0	2.10

Appendix 17
Drawing Y104L0108



Appendix 18**Materials List for Y104L0108**

Item	Component	Drawing No	Quantity	Cat No.
	Pole & Foundations			
9	Wood Pole (Stout)	Y204 L0604	2	To Suit Pole
16	Kicking / Stay Block (Type 2)	439103	4	234476
19	Brace Block (2.44m Centres)	Y204 L0301	2	217497
19	Bolts, M20 x 530 Galvd		8	372815
19	Washers, Square Curved	439605	8	368078
19	Washer, Square Flat	0183/3	8	368063
	Crossarms			
32	Crossarm Member, Conductor Plates & Stay Gusset Plates	Y204 L0904	1 Set	223669
32	Bolts, M20 x 60 bolts (Gusset Plates)		8	378703
32	Washer, Square Flat	0183/3	4	368063
32	Bolts, M20 x 300 (Med or Stout)	-	4	378794
	Insulator Assemblies			
92	*** 66kv Post Insulator	1091010487 sht 12	3	253702
60	* 66kV Tension Set	1091010488 sht 9	6	See NSP/001/027
	Sign & Notices			
70	Safety Sign	1091010229 sht1	4	363318
70 / 71	Fibre Washers, 14mm x 6mm thick		20	374581
70 / 71	Fibre Washers, 14mm x 2mm thin		20	374555
70 / 71	Brass Slotted screws No1		20	375777
71	Number base plate	1091010028 sht3	1	243258
71	Pole Numbers	1091010028 sht3	To Suit	To Suit
	Stays			
73	See NSP/001/104 Guidance document for stays Assemblies	Y204 L0501 &	4	See document for details
73	** 120kN Insulator Stay Assembly	1 09 123 0372 Sht2	4	251547
	Cable Support			
74	Anti-Climbing Guard	Y201 L1101/2	1	224519
80	66kV Sealing End Connector	Y204 L0801	3	221119
92	66kV Sealing End P-Form (15-18m)	Y204 L0909	1	227116
82	66kV Surge Diverter to be supplied with insulated base		3	224154
82	Surge Diverter Connector	Y204 L0803	3	221119
94	Cable Guard	Y204 L0911	1	224535
94	Bolts, M20 x 60 bolts		14	378703
95	Link Box	Eng Rec C55		To Suit
97	66kV Cable Supports	Y204 L0910	6	
	Earthing			
	70mm (7/3.55)HD CU Cond. PVC Covered type 8 earthing and bonding lead		20m Approx.	220146
	Comp Lug – M22 Hole (70mm PVC Cu)	1091010237	As Req	075234
	Polyprop cleats		As Req	248815
	Earthing Lead casing		1	235548
	Earth Rod	1091013143 sht 1	As Req	354249
	Saddle Clamp	1091013143 sht 2	As Req	354388

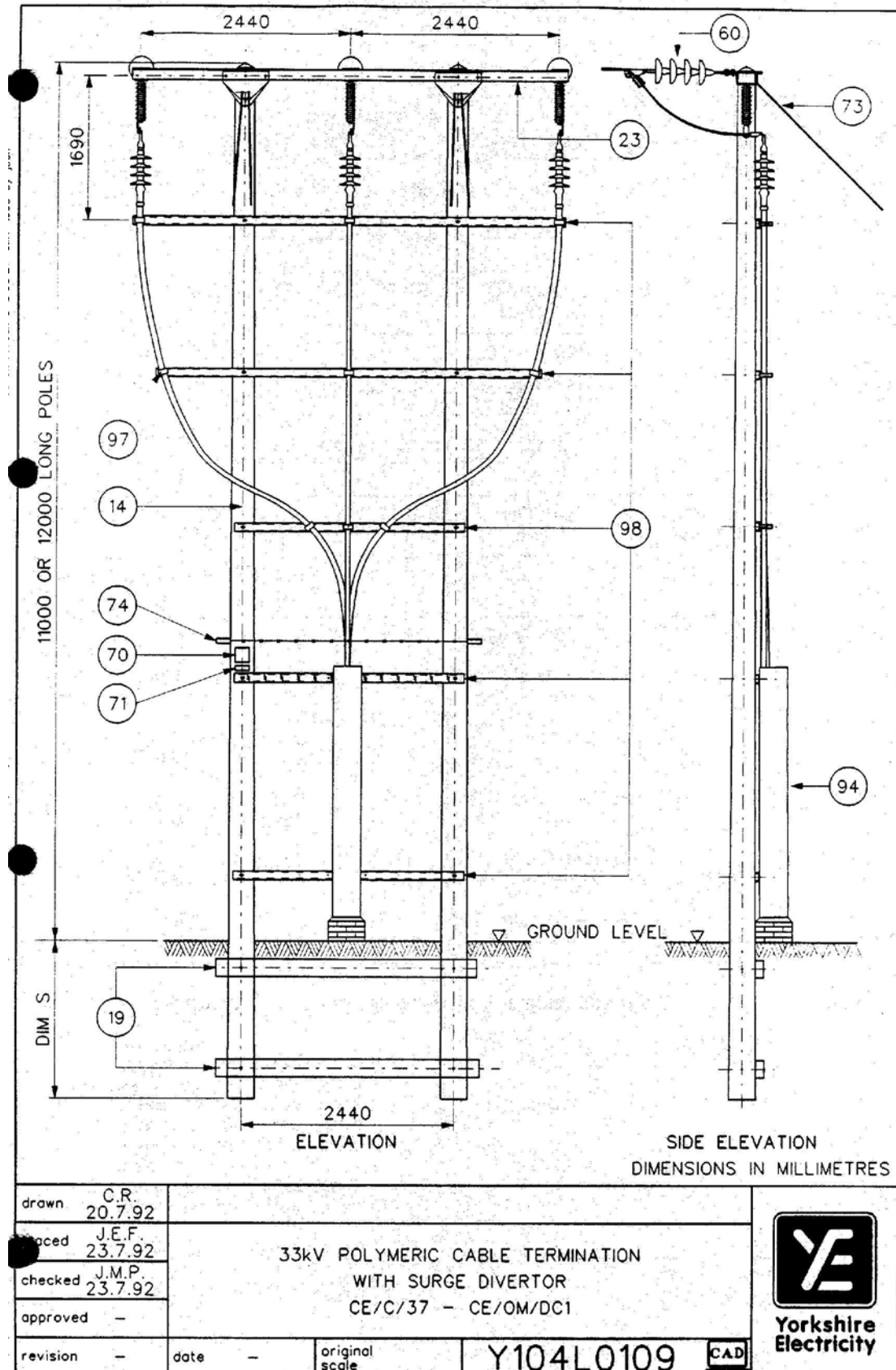
* Detailed Insulator information now available within insulator guidance document

** Updated stay assembly

*** To include insulator clamp or distribution tie to suit insulator type

Pole Planting Depth	
Pole Length (m)	Dim S (m)
15.0	2.10
16.0-17.0	2.25
18.0	2.40

Appendix 19 Drawing Y104L0109



Appendix 20**Materials List for Y104L0109**

Item	Component	Drawing No	Quantity	Cat No.
	Pole & Foundations			
14	Wood Pole (Stout)	Y204 L0605	2	To Suit Pole
19	Brace Block (2.44m Centres)	Y204 L0301	2	217497
19	Bolts, M20 x 530 Galv		4	372815
19	Washers, Square Curved	439605	4	368078
19	Washer, Square Flat	0183/3	4	368063
	Crossarms			
23	Crossarm Member, Conductor Plates & Stay Gusset Plates	Y204 L0916	1 Set	
32	Bolts, M20 x 60 bolts (Gusset Plates)		8	378703
32	Washer, Square Flat	0183/3	4	368063
32	Bolts, M20 x 300 (Med or Stout)	-	4	378794
	Insulator Assemblies			
60	* 66kV Tension Set	1091010488 sht 9	6	See NSP/001/027
	Sign & Notices			
70	Safety Sign	1091010229 sht1	2	363318
70 / 71	Fibre Washers, 14mm x 6mm thick		10	374581
70 / 71	Fibre Washers, 14mm x 2mm thin		10	374555
70 / 71	Brass Slotted screws No1		10	375777
71	Number base plate	1091010028 sht3	1	243258
71	Pole Numbers	1091010028 sht3	To Suit	To Suit
	Stays			
73	See NSP/001/104 Guidance document for stays Assemblies	Y204 L0501 &	4	See document for details
73	** 120kN Insulator Stay Assembly	1 09 123 0372 Sht2	4	251547
	Cable Support			
74	Anti-Climbing Guard	Y201 L1101/2	1	224519
98	33kV Cable support steelwork	Y204 L0907	1	
	33kV Surge Diverter	Stud / Stud Mounting	3	224113
94	Cable Guard	Y204 L0911	1	224535
94	Bolts, M20 x 60 bolts		14	378703
95	Link Box	Eng Rec C55		To Suit
97	33kV Cable Cleats		6	
97	33kv Cable Trefoil Cleats		9	
	Earthing		3	
	70mm (7/3.55)HD CU Cond. PVC Covered type 8 earthing and bonding lead		20m Approx.	220146
	Comp Lug – M22 Hole (70mm PVC Cu)	1091010237	As Req	075234
	Polyprop cleats		As Req	248815
	Earthing Lead casing		1	235548
	Earth Rod	1091013143 sht 1	As Req	354249
	Saddle Clamp	1091013143 sht 2	As Req	354388

* Detailed Insulator information now available within insulator guidance document

** Updated stay assembly

*** To include insulator clamp or distribution tie to suit insulator type

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0-13.0	2.00
14.0-15.0	2.10
16.0-17.0	2.25
18.0-20.0	2.40

Appendix 21**DESIGN SAGS AND TENSIONS 175mm².ACSR. ALL LEVELS.****CE/OM/DC1(M). 120m basic span.**

Conductor Breaking Load. 79800 Newtons.

Conductor Diameter. 19.53mm.

Conductor Weight. 0.8628 kg/m.

Conductor c.s.a. 226.2 sq.mm.

Maximum Working Tension. 24550 Newtons.

	Temp. degrees C															
Span (m)	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75
60																
65																
70																
75																
80																
85																
90																
95																
100																
105																
110																
115																
120																
125																
130																
135																
140																
145																
150																
155																
160																
165																
170																
175																
180																
	Tension at basic span															
(N)																
(kgf)																
lbf																

Appendix 22**Erection SAGS AND TENSIONS 175mm².ACSR. ALL LEVELS.
CE/OM/DC1(M). 120m basic span.**

Conductor Breaking Load. 79800 Newtons.

Conductor Diameter. 19.53mm.

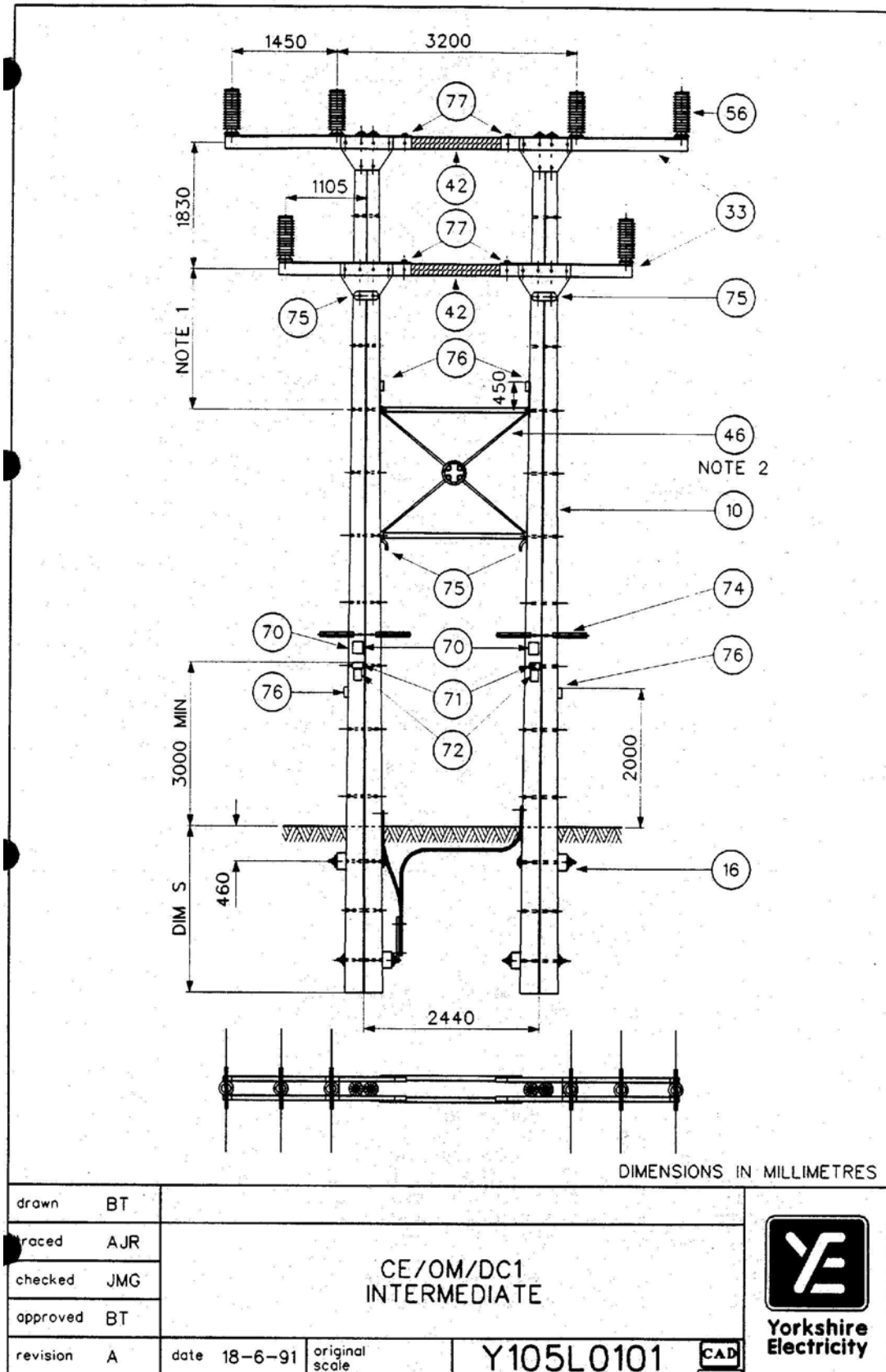
Conductor Weight. 0.8628 kg/m.

Conductor c.s.a. 226.2 sq.mm.

Maximum Working Tension. 24495 Newtons.

Span (m)	Temp. degrees C															
	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75
60	0.21	0.23	0.24	0.26	0.28	0.30	0.33	0.36	0.37	0.40	0.42	0.45	0.47	0.49	0.52	0.54
65	0.25	0.27	0.29	0.31	0.33	0.36	0.38	0.41	0.44	0.47	0.49	0.52	0.55	0.58	0.61	0.63
70	0.29	0.31	0.33	0.36	0.39	0.41	0.44	0.48	0.51	0.54	0.57	0.61	0.64	0.67	0.70	0.73
75	0.33	0.35	0.38	0.41	0.44	0.48	0.51	0.55	0.58	0.62	0.66	0.70	0.73	0.77	0.81	0.84
80	0.39	0.40	0.43	0.47	0.50	0.54	0.58	0.62	0.66	0.71	0.75	0.79	0.84	0.88	0.92	0.96
85	0.42	0.46	0.49	0.53	0.57	0.61	0.66	0.70	0.75	0.80	0.85	0.89	0.94	0.99	1.04	1.08
90	0.48	0.51	0.55	0.59	0.64	0.68	0.73	0.79	0.84	0.89	0.95	1.00	1.06	1.11	1.16	1.21
95	0.53	0.57	0.61	0.66	0.71	0.76	0.82	0.88	0.94	1.00	1.06	1.12	1.18	1.24	1.30	1.35
100	0.59	0.63	0.68	0.73	0.79	0.85	0.91	0.97	1.04	1.10	1.17	1.24	1.31	1.37	1.44	1.50
105	0.65	0.70	0.75	0.81	0.87	0.93	1.00	1.07	1.14	1.22	1.29	1.37	1.44	1.51	1.58	1.65
110	0.71	0.76	0.82	0.88	0.95	1.02	1.10	1.18	1.26	1.34	1.42	1.50	1.58	1.66	1.74	1.81
115	0.78	0.83	0.90	0.97	1.04	1.12	1.20	1.28	1.37	1.46	1.55	1.64	1.73	1.81	1.90	1.98
120	0.85	0.91	0.98	1.05	1.13	1.22	1.31	1.40	1.49	1.59	1.69	1.78	1.88	1.97	2.07	2.16
125	0.92	0.99	1.06	1.14	1.23	1.32	1.42	1.52	1.62	1.73	1.83	1.94	2.04	2.14	2.24	2.34
130	0.99	1.07	1.15	1.23	1.33	1.43	1.53	1.64	1.75	1.87	1.98	2.09	2.21	2.32	2.43	2.53
135	1.07	1.15	1.24	1.33	1.43	1.54	1.65	1.77	1.89	2.01	2.14	2.26	2.38	2.50	2.62	2.73
140	1.15	1.24	1.33	1.43	1.54	1.66	1.78	1.90	2.03	2.16	2.30	2.43	2.56	2.69	2.81	2.94
145	1.23	1.33	1.43	1.54	1.65	1.78	1.91	2.04	2.18	2.32	2.46	2.60	2.74	2.88	3.02	3.15
150	1.32	1.42	1.53	1.64	1.77	1.90	2.04	2.19	2.33	2.48	2.64	2.79	2.94	3.08	3.23	3.37
155	1.41	1.52	1.63	1.75	1.89	2.03	2.18	2.33	2.49	2.65	2.81	2.98	3.14	3.29	3.45	3.60
160	1.50	1.61	1.74	1.87	2.01	2.16	2.32	2.49	2.66	2.83	3.00	3.17	3.34	3.51	3.68	3.84
165	1.60	1.72	1.85	1.99	2.14	2.30	2.47	2.64	2.82	3.01	3.19	3.37	3.55	3.73	3.91	4.08
170	1.70	1.82	1.96	2.11	2.27	2.44	2.62	2.81	3.00	3.19	3.39	3.58	3.77	3.96	4.15	4.33
175	1.80	1.93	2.08	2.24	2.41	2.59	2.78	2.98	3.18	3.38	3.59	3.79	4.00	4.20	4.40	4.59
180	1.90	2.04	2.20	2.37	2.55	2.74	2.94	3.15	3.36	3.58	3.80	4.01	4.23	4.44	4.65	4.86
Tension at basic span																
(N)	18062	16813	15631	14522	13493	12548	11689	10915	10221	9602	9051	8561	8125	7735	7386	7072
(kgf)	1842	1714	1594	1481	1376	1279	1192	1113	1042	979	923	873	828	789	753	721
lbf	4057	3777	3511	3262	3031	2819	2626	2452	2296	2157	2033	1923	1825	1738	1659	1589

Appendix 23
Y105L0101



Appendix 24

Materials List for Y105L0101

Item	No	Commodity Code	Description	Drawing or Ref Number	M 20	Washers		
						sc	sf	r
10	1	BPC 024	Wood Pole (Type 1)	Y205 L0601				
16	4	217539	Kicking / Stay Block (Type 2)	439103	4 (Pole)	4	4	
33	1		Crossarm Member	Y205 L0901	See List			
42	4	223370	Crossarm Insert	Y205 L0904				
46	1		Pole Bracing	Y205 L0905				
56	6	215830 215863	33kV Line Post Insulator 66kV Line Post Insulator	439303 FB 2493*				
70	2	294181	Safety Sign	Y201 L1002				
71			Operational Plates	Y201 L1001/7				
72			Circuit Identification Plates	Y201 L1008				
74	2	2224519	Anti-Climbing Guard	Y201 L1101/2				
75	4		Earthing Bracket	Y205 L0906				
76	4		Flag Socket	Y205 L0908				
77	4		Danger Notice Socket	Y205 L0907				

* Doulton / Allied Reference.

Notes to Drawing No Y105 L0101

- 1520mm for poles up to 13.0 m.
2886 for poles 14.0m and above.
- Single pole bracing required for poles up to 14.0m.
Double pole bracing required for poles 15.0m and above.

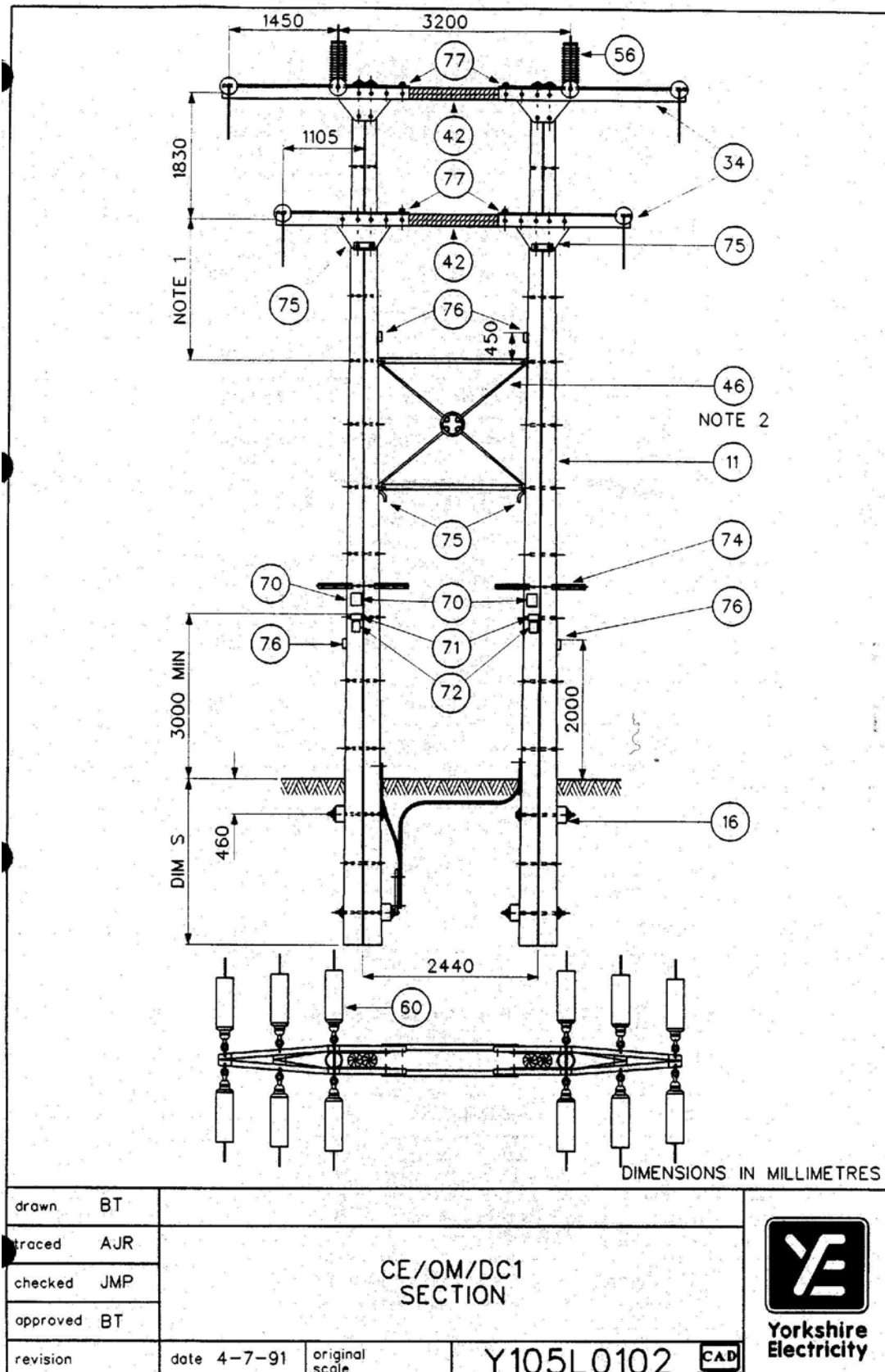
Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=14.0	2.3
>14.0	2.44

Bolt List		
Application	Bolt Detail	No
Crossarm / Pole	M20 x 260 c/w nut	8
Crossarm / Pole	M20 x 300 c/w nut	8
Crossarm / Insert	M20 x 130 ^o c/w nut	16
Crossarm / Insert	M20 x 200 ⁺ c/w nut	8
Crossarm / Gusset	M20 x 60 c/w nut	8
Crossarm / End Plate	M16 x 50 c/w nut	24

*16 off 100 x 100 x 6 square flat washers required.

+8 off 75 x 75 x 6 square flat washers required.

Appendix 25
Y105L0102



Appendix 26

Materials List for Y105L0102

Item	No	Commodity Code	Description	Drawing or Ref Number	M 20	Washers		
						sc	sf	r
11	1	BPC	Wood Pole (Type 2)	Y205 L0602				
16	4	217539	Kicking / Stay Block (Type 2)	439103	4 (Pole)	4	4	
34	1		Crossarm Member	205 L0902	See List			
42	4	223370	Crossarm Insert	Y205 L0904				
46	1		Pole Bracing	Y205 L0905				
56	2	215830 215863	33kV Line Post Insulator 66kV Line Post Insulator	439303 FB 2493*				
60	12	See NSP/001/027	* 33 or 66kV Tension Set	1091010488 sht 9				
70	2	294181	Safety Sign	Y201 L1002				
71			Operational Plates	Y201 L1001/7				
72			Circuit Identification Plates	Y201 L1008				
74	2	224519	Anti-Climbing Guard	Y201 L1101/2				
75	4		Earthing Bracket	Y205 L0906				
76	4		Flag Socket	Y205 L0908				
77	4		Danger Notice Socket	Y205 L0907				

* Doulton / Allied Reference.

Notes to Drawing No Y105 L0102

- 1520mm for poles up to 13.0 m.
2886 for poles 14.0m and above.
- Single pole bracing required for poles up to 14.0m.
Double pole bracing required for poles 15.0m and above.

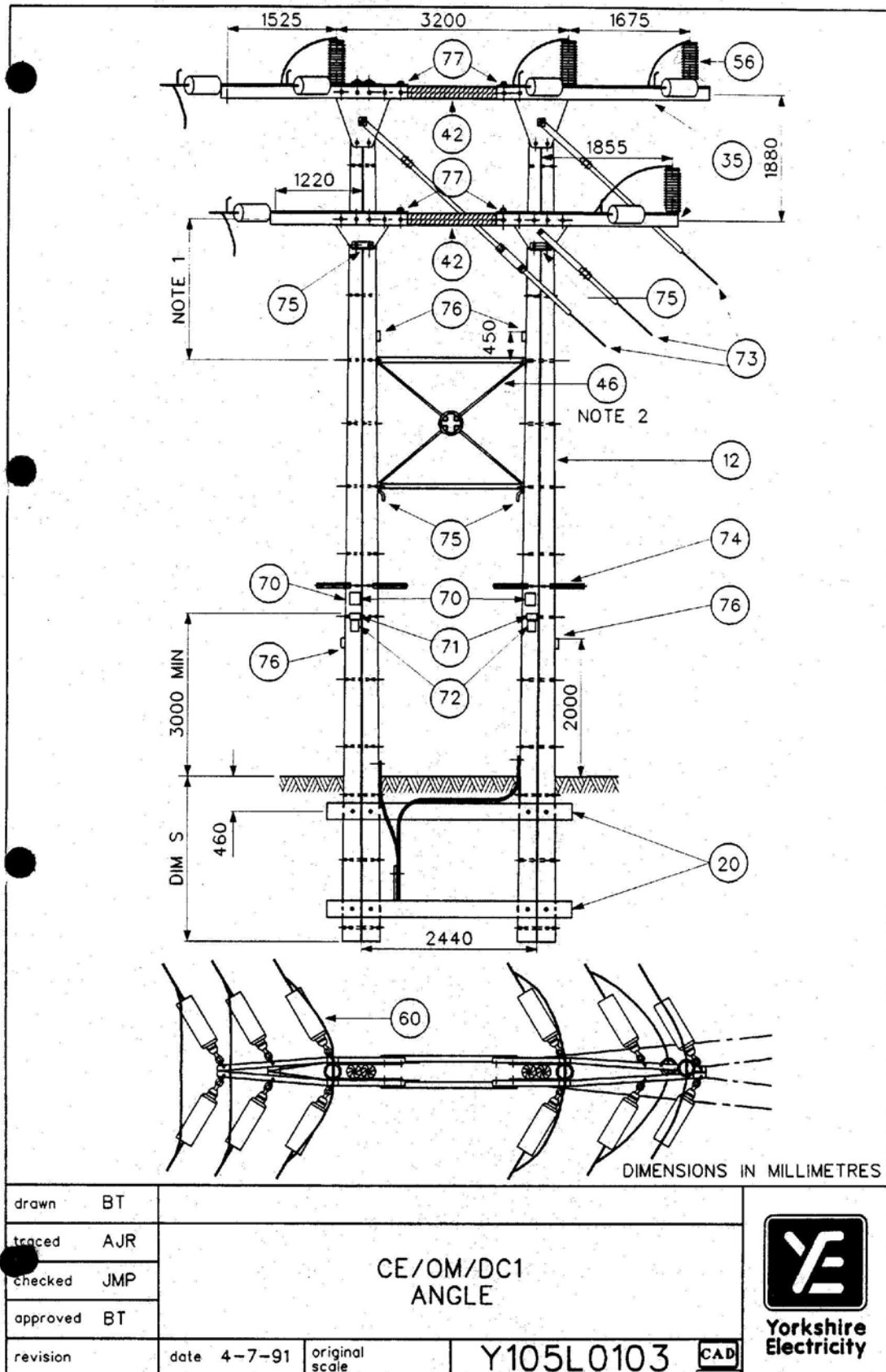
Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=14.0	2.3
>14.0	2.44

Bolt List		
Application	Bolt Detail	No
Crossarm / Pole	M20 x 300 c/w nut	16
Crossarm / Insert	M20 x 130° c/w nut	16
Crossarm / Insert	M20 x 200 ⁺ c/w nut	8
Crossarm / Gusset	M20 x 60 c/w nut	8
Crossarm / End Plate	M16 x 50 c/w nut	24

*16 off 100 x 100 x 6 square flat washers required.

⁺8 off 75 x 75 x 6 square flat washers required.

Appendix 27
Y105L0103



Appendix 28

Materials List for Y105L0103

Item	No	Commodity Code	Description	Drawing or Ref Number	M 20	Washers		
						sc	sf	r
12	1	BPC 024	Wood Pole (Type 3)	Y205 L0603				
20	2		Brace Block	Y205 L0301	4 (Pole)	4	4	
35	1		Crossarm Member	Y205 L0903	See List			
42	4	223370	Crossarm Insert	Y205 L0904				
46	1		Pole Bracing	Y205 L0905				
56	4	215830 215863	33kV Line Post Insulator 66kV Line Post Insulator	439303 FB 2493*				
60	12	See NSP/001/027	* 33 or 66kV Tension Set	1091010488 sht 9				
70	2	294181	Safety Sign	Y201 L1002				
71			Operational Plates	Y201 L1001/7				
72			Circuit Identification Plates	Y201 L1008				
73			Stay Arrangement	Y205 L0501				
74	2	224519	Anti-Climbing Guard	Y201 L1101/2				
75	4		Earthing Bracket	Y205 L0906				
76	4		Flag Socket	Y205 L0908				
77	4		Danger Notice Socket	Y205 L0907				

* Doulton / Allied Reference.

Notes to Drawing No Y105 L0103

- 1520mm for poles up to 13.0 m.
2886 for poles 14.0m and above.
- Single pole bracing required for poles up to 14.0m.

Pole Planting Depth

Pole Length	Dim S
≤14.0	2.3
>14.0	2.44

Bolt List

Application	Bolt Detail	No
Crossarm / Pole	M20 x 300 c/w nut	20
Crossarm / Insert	M20 x 130° c/w nut	16
Crossarm / Insert	M20 x 200 ⁺ c/w nut	8
Crossarm / Gusset	M20 x 60 c/w nut	8
Crossarm / End Plate	M16 x 50 c/w nut	24

* 16 off 100 x 100 x 6 square flat washers required.

⁺8 off 75 x 75 x 6 square flat washers required.